Michael Cusack-Nelkin

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Education

Columbia University | M.A. in Mathematics | New York, NY | Aug 2025 - Dec 2026

- Master's in the Mathematics of Finance (MAFN)
- GPA: 3.6 / 4.00

The College of William & Mary | B.S. in Physics | Williamsburg, VA | Aug 2018 - May 2022

- GPA: 3.7 / 4.00
- Minors in Data Science and Mathematics
- Honors: Dean's List, *magna cum laude*

Research and Competitions

High Sharpe, Low Capacity: Optimally Trading the Nasdaq Closing Auction Imbalance | Summer 2026

- Built a statistical arbitrage system combining a return forecast, 10-factor covariance model, spread costs and price impact estimates into a constrained portfolio optimization problem in Python using CVXPY.
- Backtested the strategy over 110 days and 100 stocks. Estimated an 8.53 Sharpe ratio at a maximum capacity of $91.3 million GMV.
- Identified fragility in capacity and latency requirements that favor use within a larger portfolio's execution process.

Adaptive Execution Strategies Lead to Robust Improvements Over Naive Benchmark | Spring 2026

- Co-developed an execution simulator using L2 Nasdaq order book data, supporting market, resting limit, and IOC orders under configurable latency and stochastic replenishment assumptions.
- Designed an execution strategy that used book imbalance to adjust fill aggression and reduced simulated arrival costs by 55%-89% versus a naive benchmark across three studied assets.

Millennium x Columbia University Trading Competition | New York, NY

- Led a team of four to win a Millennium Management ETF arbitrage competition for Columbia students. Used a rolling z-score to estimate a hedging relationship between the ETF and its underlying components, and traded a synthetic basket of the underlyings and ETF.

Work Experience

Deloitte Consulting | Strategy and Analytics Analyst - State of Texas | Austin, TX | Oct 2022 - Jun 2025

- Led four developers and testers in delivering a fraud recovery system for the Texas Office of Inspector General, contributing to the recovery of $10 million in previously unrecoverable fraudulent claims.
- Presented technical progress, design decisions, and testing results to a mixed audience of technical and non-technical clients multiple times per week, maintaining stakeholder support throughout development and implementation.
- Owned the project's source-of-truth technology plan and hardware distribution processes as the broader engagement expanded from approximately 30 to more than 400 team members.

DLA Piper | Financial Analyst Intern | New York, NY | Jun 2020 - Sep 2020

- Analyzed practice profitability and billing data and prepared weekly reports for finance leadership using the firm's internal financial models.

Gilder, Gagnon, Howe & Co. | Research Analyst Intern | New York, NY | Mar 2018 - Jun 2018

- Analyzed public online gaming companies using earnings calls, 10-Ks, and 10-Qs. Wrote quarterly research summaries and participated in investment committee meetings and portfolio company interviews.

Technical Skills

- Programming and Methods: Python, NumPy, Pandas, Polars, scikit-learn, statsmodels, XGBoost, CVXPY, Git, portfolio optimization, time series cross-validation, factor risk modeling, market microstructure analysis, data cleaning and normalization, technical writing with Quarto.